

Q1.

$$CMY (0.5, 0.3, 0.6)$$

$$\downarrow$$

$$RGB (0.5, 0.7, 0.4)$$

$$C_{max} = G = 0.7$$

$$C_{min} = B = 0.4$$

$$H = 2 + \frac{b - r}{C_{max} - C_{min}} = 2 + \frac{0.4 - 0.5}{0.7 - 0.4}$$

$$= 1.667 * 60$$

$$= 100$$

$$S = 1 - \frac{C_{min}}{C_{max}} = 1 - \frac{0.4}{0.7} = 0.43 \approx 43\%$$

$$V = C_{max} = 0.7 \approx 70\%$$

Q2.

$$I_A = I_a * K_a = 0.4 * 0.4 = 0.16$$

$$d_1 = \sqrt{(0-4)^2 + (8-5)^2 + (8-6)^2}$$

$$= \sqrt{29}$$

$$d_2 = \sqrt{(4-4)^2 + (2-5)^2 + (3-6)^2} = 3\sqrt{2}$$

$$f_{att1} = \max\left(1 - \left(\frac{d}{n}\right)^2, 0\right)$$

$$= \max\left(1 - \left(\frac{\sqrt{29}}{5}\right)^2, 0\right)$$

$$= 0$$

$$f_{att2} = \max\left(1 - \left(\frac{3\sqrt{2}}{5}\right)^2, 0\right)$$

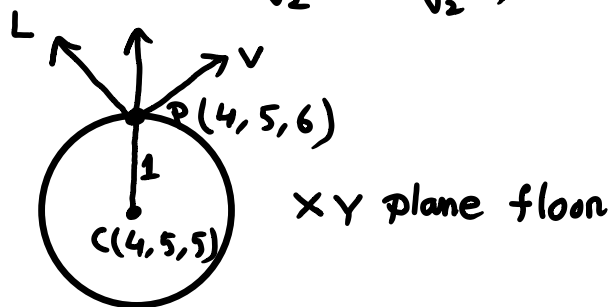
$$= 0.28$$

As $f_{att1} = 0$, we don't need to calculate diffuse and specular light for the first light.

For 2nd light,

$$\vec{L} = L_2 - P = (4, 2, 3) - (4, 5, 6) = (0, -3, -3)$$

$$\hat{L} = (0, -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$$



$$\vec{N} = P - C = (4, 5, 6) - (4, 5, 5) \\ = (0, 0, 1)$$

$$\hat{N} = (0, 0, 1)$$

$$\hat{L} \cdot \hat{N} = -\frac{1}{\sqrt{2}} \rightarrow \text{Negative } (-), \text{ So, } I_{d2} = 0$$

$$\hat{R} = 2(\hat{L} \cdot \hat{N})\hat{N} - \hat{L} = (0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$$

$$\vec{V} = V - P = (2, -2, 1)$$

$$\hat{V} = \frac{2}{3}, \frac{-2}{3}, \frac{1}{3}$$

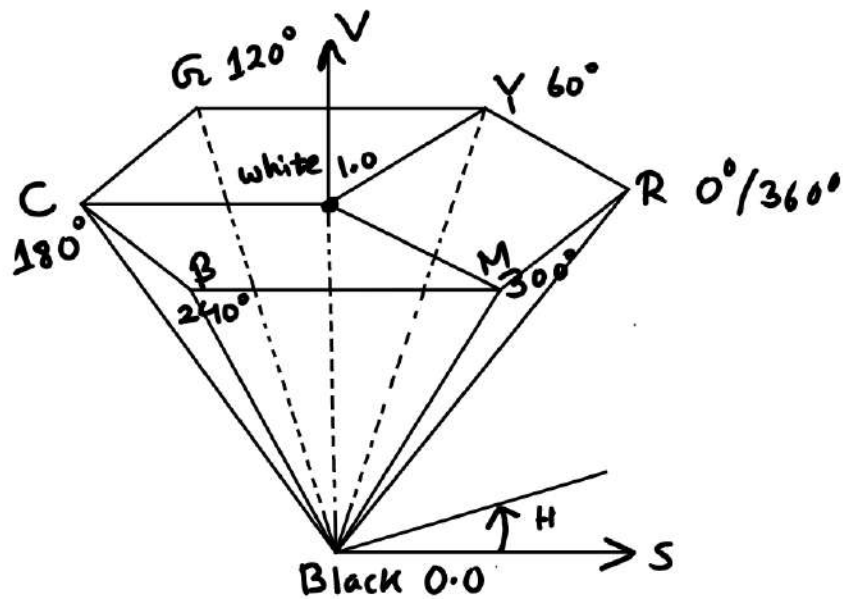
$$\hat{R} \cdot \hat{V} = -\frac{1}{\sqrt{2}} \rightarrow \text{Negative } (-), \text{ So, } I_{s2} = 0$$

$$I = I_A + \sum_{i=1}^2 f_{att_i} (I_{d_i} + I_{s_i})$$

$$= I_A + 0 = I_A = 0.16 \quad (\underline{\text{Ans}})$$

Q4. [Additional]

(a)



b) CMY (0.1, 0.2, 0.3)

↓
RGB (0.9, 0.8, 0.7)

$$dx = 423 - (-423) \\ = 846$$

$$dy = 370 - 423 \\ = -53$$

$$|dx| > |dy|$$

$$\text{Color}_D = A + \frac{d_{AD}}{d_{AC}} (C - A)$$

$$= (0.9, 0.8, 0.7) + \frac{0+423}{423+423} (0.1-0.9, 0.2-0.8, 0.3-0.7)$$

$$= (0.5, 0.5, 0.5) \quad (\underline{\underline{\text{Ans}}})$$